HARDWARE TROUBLESHOOTING GUIDE

(Beginner-Friendly • Step-by-Step • Chatbot-Optimized)

GENERAL CHATBOT DESIGN NOTE (IMPORTANT)

This guide is written so a chatbot can:

- Jump to **exact procedures** when user asks *how to do something*
- Answer **“how”, “where”, “why”, “steps”, “fix”** questions
- Avoid giving unsafe advice unless clearly marked

Each device contains:

- Problems
- Causes
- **Very detailed solutions**
- **Chatbot trigger phrases** (for intent recognition)

DEVICE: LAPTOP

PROBLEM 1: LAPTOP POWERS ON BUT NO DISPLAY

Symptoms

- Power light ON

- Fan spinning
- Black screen
- No logo

Possible Causes

- Loose RAM
- Faulty RAM
- Display output stuck
- Loose display cable

SOLUTION A: RESEAT RAM (BEGINNER STEP-BY-STEP)

⚠️ Safe for normal users if followed carefully

Steps:

1. Shut down the laptop completely
2. Unplug the charger
3. If battery is removable, remove it
4. Hold power button for **10 seconds** (discharge electricity)
5. Place laptop upside down on a flat table
6. Remove all back-panel screws
7. Gently lift the back panel (do NOT force)
8. Locate RAM
    - Small rectangular chip

    - Held by **two metal clips** on sides
9. Push both clips outward
10. RAM will pop up at an angle
11. Hold RAM by edges and pull out
12. Reinsert RAM at **30–45° angle**
13. Press down until clips click
14. Close back panel and tighten screws
15. Reconnect battery & charger
16. Power on and check display

CHATBOT TRIGGERS

- “How to remove RAM”
- “Laptop RAM steps”
- “Laptop black screen but power on”
- “Where is RAM in laptop”

PROBLEM 2: LAPTOP OVERHEATING

Causes

- Dust blocking vents
- Fan obstruction
- Soft surface usage (bed, pillow)

SOLUTION (SAFE)

1. Turn off laptop
2. Unplug charger
3. Use compressed air on vents
4. Keep laptop on hard surface
5. Avoid using on bed
6. Use cooling pad if available

CHATBOT TRIGGERS

- “Laptop is overheating
- “Laptop too hot”
- “How to fix laptop overheating”

PROBLEM 3: LAPTOP NOT CHARGING

Causes

- Charger issue
- Charging port dust
- Battery wear

SOLUTION

1. Try another charger
2. Check power socket
3. Gently clean charging port (wooden toothpick)

4. Restart laptop
5. If still not charging → battery likely weak

CHATBOT TRIGGERS

- “Laptop is not charging”
- “How to check if Laptop battery is faulty or weak”
- “Laptop not charging”

DEVICE: TELEVISION (TV)

PROBLEM 1: TV NOT TURNING ON

Symptoms

- No power light
- No sound
- No display

Causes

- No power supply
- Faulty remote
- Internal power board

SOLUTION (USER-LEVEL)

1. Check wall socket with another device

2. Replug TV power cable firmly
3. Try turning on via **TV button**, not remote
4. Replace remote batteries
5. Power reset:
    - Unplug TV
    - Wait 5 minutes
    - Plug back and try

CHATBOT TRIGGERS

- “TV not turning on”
- “TV no power light”
- “TV dead”

PROBLEM 2: SOUND BUT NO PICTURE

Causes

- Backlight failure
- Display settings issue

SOLUTION (SAFE CHECKS)

1. Shine flashlight on screen
2. If faint image visible → backlight issue

3. Increase brightness from settings
4. Factory reset TV if menu visible

⚠️ Backlight replacement = technician job

CHATBOT TRIGGERS

- “TV screen is not visible”
- “TV screen is blank but sound is coming”
- “How to replace TV backlight”
- “How to check if TV has a backlight issue”

PROBLEM 3: NO SOUND BUT PICTURE OK

SOLUTION

1. Increase volume
2. Check mute setting
3. Disconnect headphones
4. Change audio output settings
5. Try external speakers

CHATBOT TRIGGERS

- “No sound in TV”
- “TV Display is one but no sound”
- “How to fix no sound problem”

DEVICE: AIR CONDITIONER (AC)

PROBLEM 1: AC NOT COOLING

Symptoms

- Air blowing but not cold

Causes

- Dirty filter
- Incorrect mode
- Low gas

SOLUTION A: CLEAN FILTER (USER SAFE)

1. Turn off AC
2. Open front panel
3. Remove air filters
4. Wash with water
5. Dry completely
6. Reinstall
7. Turn on AC

SOLUTION B: CHECK SETTINGS

1. Set mode to **Cool**

2. Temperature ≤ 24°C
3. Fan speed medium/high

CHATBOT TRIGGERS

- “AC not cooling”
- “AC blowing air but not cold”
- “How to clean AC filter”

PROBLEM 2: WATER DRIPPING FROM AC

Causes

- Blocked drain pipe

SOLUTION

1. Turn off AC
2. Locate drain pipe outside
3. Blow air gently or flush with water
4. Ensure pipe slopes downward

CHATBOT TRIGGERS

- “How to fix a blocked drain pipe.”
- “How to fix water dripping from AC”
- “AC water leakage problem”

DEVICE: MOBILE PHONE

PROBLEM 1: PHONE NOT TURNING ON

Causes

- Battery fully drained
- Software crash

SOLUTION

1. Charge phone for 30 minutes
2. Try long press power button (15 seconds)
3. Try power + volume button
4. Use original charger if possible

CHATBOT TRIGGERS

- “Phone not turning on”
- “Mobile dead”
- “Phone won’t start”

PROBLEM 2: PHONE OVERHEATING

Causes

- Heavy apps
- Charging while using
- Background apps

SOLUTION

1. Close unused apps
2. Remove phone cover
3. Stop charging temporarily
4. Restart phone
5. Avoid gaming while charging

CHATBOT TRIGGERS

- “Phone is overheating”
- “Mobile is too hot”
- “How to fix mobile’s overheating”

PROBLEM 3: PHONE NOT CHARGING

SOLUTION

1. Try different cable
2. Clean charging port (wooden toothpick only)
3. Try different socket
4. Restart phone

CHATBOT TRIGGERS

- “Phone charging issue”
- “Charging cable not working”
- “Mobile phone not charging”

DEVICE: CEILING FAN

PROBLEM 1: FAN NOT TURNING ON

Causes

- No power
- Faulty switch
- Capacitor issue

SOLUTION (SAFE)

1. Check wall switch
2. Test power using light/fan tester
3. Turn regulator knob
4. If humming but not spinning → capacitor likely weak

CHATBOT TRIGGERS

- “Fan not working”

- “Fan not starting”

- “Fan humming but not spinning”
- “Sound coming but Fan is not spinning”
- “Fan is not turning on”
- “Fan button not working”

PROBLEM 2: FAN RUNNING SLOWLY

Causes

- Weak capacitor

- Dust in bearings

- Low voltage

SOLUTION

1. Clean fan blades

2. Oil bearings (if accessible)

3. Replace capacitor (electrician recommended)

CHATBOT TRIGGERS

- “Fan is on but no air”

- “Fan is very slow”

- “Fan running very slow”